

A Novel Sensor for Simultaneous Measurement of Force and Position

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Abstract—This study introduces a novel sensor which is capable of providing simultaneous measurements of force and position. This sensor has potential application in automation and robotics. A major advantage of the force-position sensor is that both force and position quantities are obtained as continuous, not discrete values. The sensor is based on the deflection of a cantilever beam. When a point force is applied to the cantilever, multiple sets of strain gauges are used to obtain information about the magnitude of the applied force and its position along the beam. Three different designs for the sensor concept were designed, tested, and validated. The results indicate that the force-position sensor is highly linear and accurate, with the fit of measured positions, forces and angles to true applied values having an R^2 of over 0.98. The parameters that influence performance, design considerations, potential applications, and limitations of the force-position sensor are addressed.

Index Terms— force sensor, position sensor, tactile sensor, load cell

I. Introduction

FORCE and position sensing are fundamental problems in the field of engineering measurement. Treated alone, these two problems are well defined and have many robust solutions, but sensor solutions are much more limited for simultaneously measuring both the magnitude of an applied force and its position.

Load cells are sensors used to measure an applied force or torque and are used in many industrial and research applications [1]. Reference [2] gives an excellent overview of the various mechanical means of measuring force. The most common approach uses a mechanical structure that converts the applied force to a bending strain. The resulting strain is then measured with one or more strain gauges. This approach is used for two reasons. First, bending tends to amplify strain levels. Second, bending strains are highly amenable to temperature compensation [3]. Common strain-gauge based load cell structures include simple cantilevers, parallel plate cantilevers, S-beams, rings, and spheres [4], [5], [6].

Measurement of applied forces and moments with components in multiple axes requires more complex load cell designs with more sophisticated strain gauge configurations. Examples of this include [7] that used a wheel-and-spoke structure to measure 6-axis forces, and [8] that used a G-shaped sensor to measure forces with magnitudes in two distinct ranges.

In virtually all cases, load cells require that the force being measured is applied at a specified position. Inaccurate measurements will result if forces are applied at other positions on the load cell structure or if the load direction is not properly aligned with the load cell. Traditional load cells thus require that the force position be closely controlled.

The problem of measuring an applied force with a variable

position has been most thoroughly explored in the field of tactile sensing. Tactile sensing addresses the sensing of mechanical stimuli such as pressure, strain, and shear, and environmental stimuli such as humidity and temperature [9], [10]. Tactile sensing is often accomplished using thin, flexible arrays of individual micromechanical sensors. Positional data is typically gathered by analyzing which sensors in the array are stimulated. This approach provides only discrete values for position.

Common sensing mechanisms for the sensors in tactile arrays include capacitive sensing, where an external force changes the conductivity of a dielectric at the point of contact [11], [12]; piezoelectric sensing, which involves measuring an increase in stored charge within specific substrates as a result of applied forces [13], [14]; and piezoresistive sensing, where a mechanical input changes the resistivity of a sensing substrate [15], [16]. Simpler, low-fidelity forms of tactile sensors are simply made of arrays of small load cells [12], [17], [18] or single strain gauges placed strategically on deformable surfaces to detect any change in shape [19], [20], whereas high fidelity sensors can be made of large, dense arrays of small taxels that can sense at high spatial resolutions [21], [22].

One drawback to tactile sensors is their lack of robustness to abrasion and wear, which can damage the sensing structure and reduce the lifespan of the device. Another limitation is their relatively low force ranges; many tactile sensors are designed to measure contact forces of less than 10 N [23], [24].

In this paper, we introduce a novel sensor designed to simultaneously measure the magnitude of an applied force and the position of that force along a beam. Inspired by the rugged and simple nature of strain gauge-based load cells, this sensor relies upon at least two strain gauge Wheatstone bridges mounted on a cantilever beam. By combining information from

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both strain gauges, both the applied force and its position along the beam are determined. Design principles are discussed that make the sensor concept adaptable to a variety of applications.

Few attempts have been made at solving the force-position sensing problem in this way. A similar concept was introduced for a biomedical application in [25], where thick-film piezoresistors in three bridges were used to detect the position and magnitude of a force on a 2-dimensional plate. However, few technical details were provided, and the design principles were not addressed. [26] uses an array of digital optical proximity sensors embedded in an elastomer to detect contact forces and position along a beam for use in a robotic gripper, but the contact forces are limited to 5 N. The elastomer coating, while suitable for gripping, is susceptible to wear and abrasion, which may affect the optical properties of the sensor.

This paper will be organized as follows. In the Methods section we derive the force-position sensor governing equations for measurement and calibration and introduce three prototype designs that were built and tested. The Results section shows the calibration results and measurement accuracy of each prototype. In the Discussion section we explore design parameters and their effect on the sensor's performance. We conclude with general guidance for making design decisions. We will also discuss the sensor's limitations and potential applications.

II. METHODS

A. Overview

Like many load cells, our sensor utilizes resistive foil strain gauges. However, whereas most load cells are designed to measure force at a specified position, our concept uses two (or more) strain gauges to simultaneously provide measurement of force and the position of that force.

In its simplest implementation, the sensor consists of two strain gauges bonded to a cantilever beam. As shown in Fig. 1, these two strain gauges must be located at distinct axial locations. Each strain gauge is incorporated into a separate quarter-bridge Wheatstone bridge circuit. Additional strain gauges can be mounted at each location to form half-bridge or full bridge configurations. For the remainder of this paper, we will use the term *strain gauge* to refer to one or more strain gauges in a Wheatstone bridge configuration.

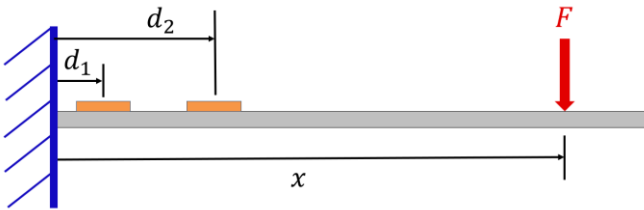


Fig. 1. Diagram of a sensor consisting of a cantilever beam with strain gauges bonded at two locations. The strain gauge positions, force position and force magnitude are defined with variables.

B. Principle of Operation

For small displacements, the bending strain ε of a cantilever beam is linearly related to the bending moment:

$$\varepsilon = \frac{My}{EI} \quad (1)$$

Here, E is the Young's modulus of the beam, I is the beam's moment of inertia, and y is the distance from the point of force application to the neutral axis.

When a force F is applied at a distance x from the cantilever's base, the resulting bending moment experienced by a strain gauge i mounted on the beam's surface a distance d from the base is given by:

$$M_i = F(x - d_i) \quad (2)$$

Thus, the expression for the strain experienced by different strain gauges is given as

$$\varepsilon_i = \frac{y}{EI} F(x - d_i) \quad (3)$$

As is typical, the strain gauge is read using an amplified Wheatstone bridge circuit. The equation for the voltage produced by the Wheatstone bridge is a function of the strain:

$$V_i = \frac{V_s G g n}{4} \varepsilon_i \quad (4)$$

Here V_s is the supply voltage, G is the gauge factor of the strain gauge, g is the amplification (gain), and n is a variable relating to the Wheatstone bridge configuration. For quarter, half, and full Wheatstone bridge configurations, n equals 1, 2, and 4, respectively [3]. Combining (3) and (4), we obtain an expression for the voltage in terms of the input force and position:

$$V_i = \frac{V_s G g n y}{4EI} F(x - d_i) \quad (5)$$

The above equation represents the measured voltage in an ideal scenario. In practice, an imperfectly balanced Wheatstone bridge can be modeled using a modified form of this equation:

$$V_i = k_i F(x - d_i) + c_i \quad (6)$$

Here, c represents bias caused by imbalance in the Wheatstone bridge and k accounts for all the constant terms from (4):

$$k = \frac{V_s G g n y}{4EI} \quad (7)$$

In typical load cells, the force F is unknown, and the term related to application distance $(x - d)$ is fixed and thus becomes absorbed in the constant, k .

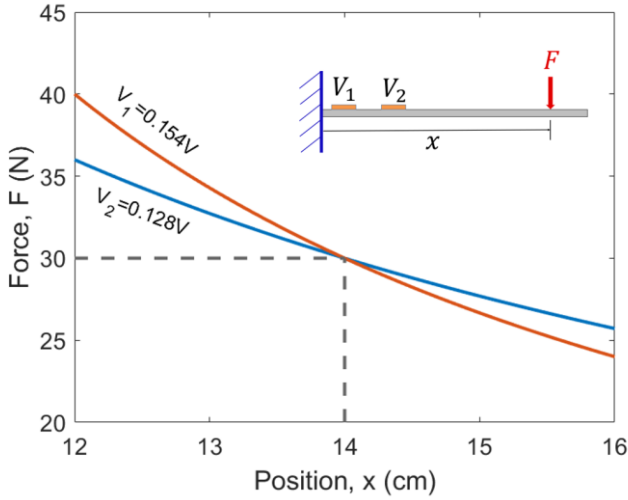


Fig. 2. Voltage isolines for two strain gauges. The intersection of the voltage lines represents the actual applied force, F and position where the force is applied, x .

The new sensor described in this study makes use of at least two strain gauges mounted on the beam at locations d_1 and d_2 . When a single force F is applied at a distance x from the base, the output voltages from the two strain gauges are:

$$V_1 = k_1 F (x - d_1) + c_1 \quad (8)$$

$$V_2 = k_2 F (x - d_2) + c_2 \quad (9)$$

Fig. 2 illustrates the measurement process graphically. Here, the voltage experienced by each strain gauge is plotted as a voltage isoline on a distance-force plot, representing all the combinations of F and x that could generate that voltage. The point (x, F) where the two isolines intersect represents the solution to the system of equations, the only combination of x and F that could produce both voltages. Note that at this point the voltages are not equal - voltage does not change along the figure axes. Voltage isolines will be explored further in the discussion section.

In practice, (8) and (9) can be solved algebraically yielding the following equations:

$$x = \frac{k_2 d_2 (V_1 - c_1) - k_1 d_1 (V_2 - c_2)}{k_2 (V_1 - c_1) - k_1 (V_2 - c_2)} \quad (10)$$

$$F = \frac{k_2 (V_1 - c_1) - k_1 (V_2 - c_2)}{k_1 k_2 (d_2 - d_1)} \quad (11)$$

C. Calibration & Prediction

In theory, the k_i values could be estimated from individual measurements of all the variables in (7). However, it is both unnecessary and impractical to directly measure all these variables. A better approach is to obtain k_i values through empirical calibration. The process described below provides an efficient method for obtaining reliable estimates of c_i and d_i

values through calibration rather than through direct measurement of the constituent elements.

We start by rearranging (8) and (9) as follows:

$$V_i = k_i (Fx) - k_i d_i (F) + c_i \quad (12)$$

Using a simple change of variables $Fx \rightarrow z_1$, $F \rightarrow z_2$, the equation above is converted into the standard form for multiple linear regression:

$$V_i = \beta_{0,i} + \beta_{1,i} (z_1) + \beta_{2,i} (z_2) \quad (13)$$

With three unknown constants in (13), a minimum of three measurements are required to perform calibration for each strain gauge. Calibration is performed using calibrated masses which are placed at known positions on the sensor. The use of three points will typically not span all the desired values of F and x . A superior minimum calibration scheme is to choose two levels for F and two levels for x . This results in a total of 4 measurements, but additional points can also be used. Least squares regression is then used to solve for the unknown coefficients in (13). The individual calibration constants are then obtained as:

$$c_i = \beta_{0,i} \quad (14)$$

$$k_i = \beta_{1,i} \quad (15)$$

$$d_i = \beta_{2,i} / \beta_{1,i} \quad (16)$$

One set of 4 calibration tests is sufficient to calibrate both strain gauges if voltages from both strain gauges are recorded simultaneously. Once the values of k_i , d_i and c_i have been obtained for each strain gauge, measurements of applied forces and distances are possible based upon recorded voltages from the two strain gauges.

TABLE I
PROTOTYPE PHYSICAL DIMENSIONS

Prototype	A	B	C
Beam Profile	Rectangular	Rectangular	Tubular
Material	6061 Alum.	2024 Alum.	6061 Alum.
Length (cm)	80	45	20
Width/Diameter (cm)	5	1.27	1.27
Wall Thickness (cm)	0.318	1.27	0.165
Bridge 1 Position (cm)	2	2	2
Bridge 2 Position (cm)	12	10	7



Fig. 3. CAD models of Prototypes A (top right) and B (bottom left). Strain gauges are shown in orange.

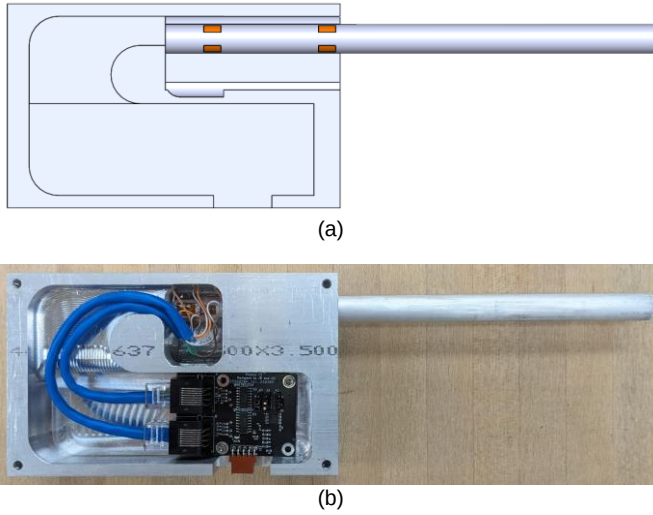


Fig. 4. CAD model (a) and physical construction (b) of Prototype C. The prototype protects the strain gauges within a cavity in the mounting block, so the CAD model shows a section view to highlight the location of the strain gauges.

D. Prototype Construction

To validate the sensor concept, three different prototype designs were constructed. Detailed dimensions for each prototype are included in Table I.

Prototype A was constructed using a rectangular aluminum bar. Strain gauges were mounted on the top and bottom of the bar, left and right of the centerline, for a total of 8 strain gauges wired into two full bridges.

Prototype B consisted of a square bar of aluminum with strain gauges mounted on the top and bottom of the beam in half bridges. At each strain gauge location, a 0.95 cm diameter hole was drilled through the beam. These holes act to amplify the strain at the location of the strain gauges. The use of such holes is common in load cell designs.

Prototypes A and B used cantilever beams of rectangular cross-sections which are appropriate for measuring a force acting perpendicular to the plane in which the strain gauges are mounted. A hollow, circular aluminum rod was used in Prototype C with two sets of strain gauges. This configuration enabled the detection of oblique forces.

E. Prototype Calibration & Prediction

1) Data Collection

A data set was generated for each prototype by performing a series of measurements using known forces applied at known positions. Forces were applied to each prototype using a set of calibrated masses (0.25, 0.5, 1, 1.5, and 2 kg). The beam of each prototype was oriented horizontally, and masses were hung vertically from the sensor at specified distances along the beam's length.

For Prototype A, each weight was hung at positions from 15 to 55 cm in 5 cm increments. Prototype B used positions of 15 to 45 cm in 5 cm increments. Prototype C used positions of 10 to 20 cm in 2 cm increments. For Prototype C, weights were hung with the device tilted at angles from 0 to 90 degrees, in 15-degree increments.

For each calibration force/position pair, the voltage from each strain gauge was digitally recorded using standard electronic measurement techniques.

2) Calibration Data

Two subsets of the collected data were used to perform calibration. In the first subset, four force/position data were reserved for this purpose. These four points corresponded to the 2x2 grid of highest and lowest forces and positions. In the second subset, a 3x3 calibration grid was used for a total of nine calibration data points. In both cases, calibration parameters k , d , and c were obtained for each prototype using the calibration process described above.

TABLE II
CALIBRATION R^2 VALUES

Prototype	A	B	C_y	C_z
Bridge Type	Full	Half	Half	Half
Bridge 1 (4 pts)	0.99990	1.00000	0.99997	0.99993
Bridge 1 (9 pts)	0.99942	0.99999	0.99980	0.99966
Bridge 2 (4 pts)	0.99984	1.00000	0.99999	0.99996
Bridge 2 (9 pts)	0.99917	0.99992	0.99976	0.99974

3) Prediction & Validation

The remaining data (all data not used for calibration) were used to assess the predictive accuracy of each prototype. However, in the validation stage, recorded voltages were used in conjunction with (10) and (11) to predict the corresponding applied force and its position. Predictions were compared to the true applied force and distance to assess the accuracy of each prototype.

Prototypes A and B each have a single pair of strain gauge bridges, resulting in one set of measurements for force and position. For prototype C, two pairs of strain gauge bridges were used in perpendicular axes, resulting in two measurements of force and position in two different axes. The two force measurements provide two components of the applied force, which we use to calculate the magnitude F and angle θ the force is applied in the plane perpendicular to the axis of the cantilever beam. The two position measurements are combined using a weighted average based on this predicted angle to determine the position x of the applied force.

TABLE III
PREDICTION R^2 VALUES

Prototype	A	B	C_y
Position	0.998	0.987	0.992
Force	1.000	0.999	0.999
Angle	N/A	N/A	0.997

III. RESULTS

A. Calibration Results

Calibration quality was assessed by computing the R^2 value for each prototype's calibration. For each prototype, calibration was first conducted using a 2x2 grid of four calibration points spanning the sensor's force and distance measurement regions, then using a 3x3 grid of 9 calibration points. Overall, the calibration process exhibited very high R^2 values with a median value of 0.9999 and a minimum value of 0.99917. The individual R^2 values are reported in Table II.

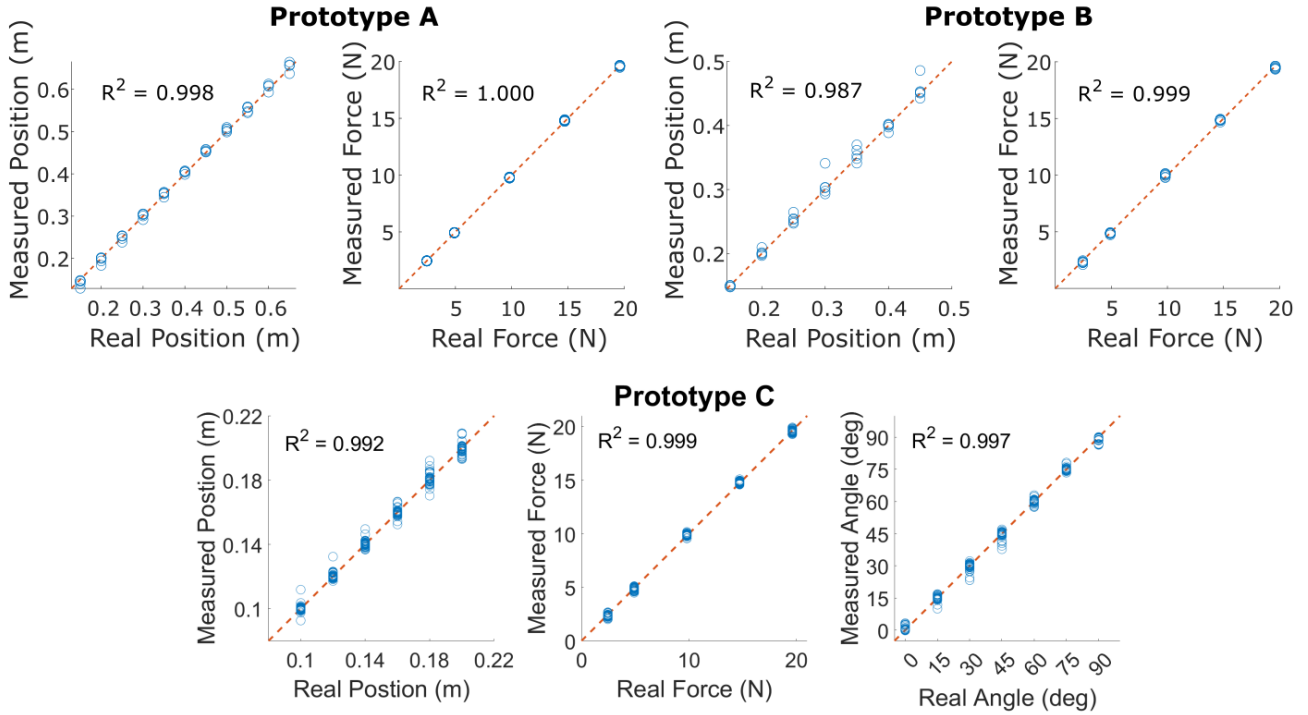


Fig. 5. Position, force, and angle measurements vs. actual values for prototypes A, B, and C. The red line indicates a 1:1 line between predicted and true. Measurements closer to this line are more accurate. Each plot displays the R^2 value for the fit of the data to this 1:1 line. Prototype A, B, and C have sample sizes of $n = 53$, 35 , and 210 , respectively.

B. Prediction Results

For each prototype, predictions for F and x (and θ for Prototype 3) were assessed by plotting the predicted values against the corresponding true values. For a perfect sensor, all predictions would lie on a line with slope 1, indicating that the prediction corresponds perfectly with the true applied value. To evaluate the accuracy of the prototypes, the R^2 value of the prediction data points with this 1:1 line was calculated for each measurement. These results are shown in Fig. 5 and Table III.

IV. DISCUSSION

In this section, we will address the influence of design parameters on the performance of the force-position sensor. As will be shown, the critical design parameters that affect performance include strain gauge spacing, the length of the measurement region, and geometric and material properties of the beam. We will also address options for calibration point selection.

A. Maximum and Minimum Moment

The measurement ranges of the force-position sensor are defined by the minimum moment the sensor can detect and the maximum moment the device can withstand before plastic deformation occurs.

1) Maximum Moment

The maximum moment the sensor can detect is bounded by the elastic limit of the measurement beam. If a sufficiently large moment is applied, the beam will deform plastically, and the sensor will need to be re-calibrated or replaced. For metals,

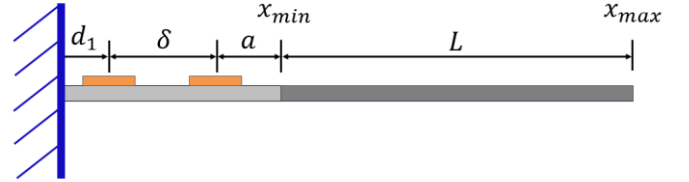


Fig. 6. The length of the cantilever beam is broken into different quantities for analysis.

plastic deformation typically will not occur for strains below 0.1%, but the exact level at which plastic deformation will occur depends upon the specific alloy being used. Regardless of alloy, the maximum bending strain will occur at the base of the cantilever when the largest allowable force is applied at the end of the cantilever:

$$\epsilon_{max} = \frac{F_{max} x_{max} y}{3EI} \quad (17)$$

Here, y is the distance between the top surface of the beam and the neutral axis. The sensor material will determine the maximum strain, ϵ_{max} and the elastic modulus, E . Once these values are known, (17) can be used to assess tradeoffs between F_{max} , x_{max} , and the cross-sectional parameters of the beam that produce the cross-sectional moment of inertia, I .

The maximum strain detectable by the sensor will correspond to the moment experienced by strain gauge 1 when F_{max} is applied at x_{max} :

$$M_{max} = F_{max} (x_{max} - d_1) \quad (18)$$

2) Minimum Moment

Theoretically, there is nothing preventing the force-position sensor system from measuring an infinitesimally small force. However, in practice, detectability will be limited by the resolution and noise characteristics of the electronic measurement system. The smallest voltage measured by the system will originate from strain gauge 2 when small forces are applied at positions very close to the gauge. The signal-to-noise ratio imposes a measurement floor; when the voltage measured from gauge 2 drops below a certain level, the measurement can no longer be considered reliable.

The sensor's electronics should be designed so that the smallest desired load, F_{min} applied at the desired start of the measurement region, x_{min} produces a voltage from gauge 2 that is above the noise floor. The noise floor can be assessed in the calibration process by measuring the standard deviation of the ambient voltage read by the strain gauges when no load is applied. The equation for the minimum voltage that can be measured is found by substituting minimum values into (5):

$$V_{min} = k_2 F_{min} (x_{min} - d_2) + c_2 \quad (19)$$

The term $F_{min}(x_{min} - d_2) + c_2$ is equivalent to the minimum moment that can be measured:

$$M_{min} = F_{min} (x_{min} - d_2) \quad (20)$$

3) Selecting Calibration Points

The maximum and minimum moments introduced above can be visualized as isolines on a force-moment diagram. The isoline for maximum moment defines a line above which the sensor will be damaged by plastic deformation. The minimum moment isoline provides the boundary below which the signal from strain gauge 2 is so low that measurement becomes undetectable or unreliable. As shown in Fig. 7, these lines define the region of all possible measurements that can be collected with this sensor.

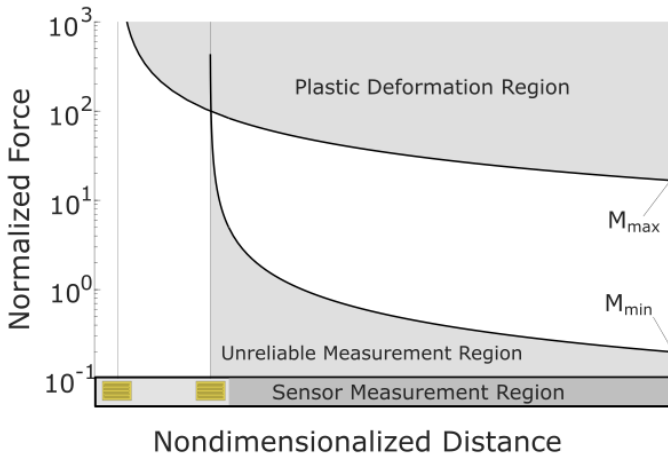


Fig. 7. Moment isolines (curves) with regions of plastic deformation and unreliable signal notated in gray.

As discussed previously, each strain gauge requires a minimum of three force/position calibration points. These three points allow (8) and (9) to be solved for the three calibration

constants. However, because these equations will generally be used to infer ranges of both force and position values, a minimum of 4 points are needed to obtain a set of calibration constants that does not rely upon extrapolation. As with all sensors, accuracy and reliability will increase as more sample points are included in the calibration process.

Fig. 8 shows the minimum recommended choice for four calibration points in black. Additional calibration points are represented by white circles. The corresponding measurement range is shown in blue. Although measurements are feasible within the entire measurement region, the practical measurement region is restricted to the blue rectangular region, which is bounded above and below by maximum and minimum design forces. Performance constraints limit the practical usability of the sensor above and below these force values. Forces above the maximum design force can be measured, but such forces may cause plastic deformation which will damage the sensor. Similarly, forces lower than the minimum design force can be measured, but depending on the position of the force, such forces may be undetectable.

Additional calibration points can be selected within the measurement region or on its edges if desired. However, the results above showed strong linearity in their calibration, indicating that additional calibration points will not have a large effect on the overall accuracy.

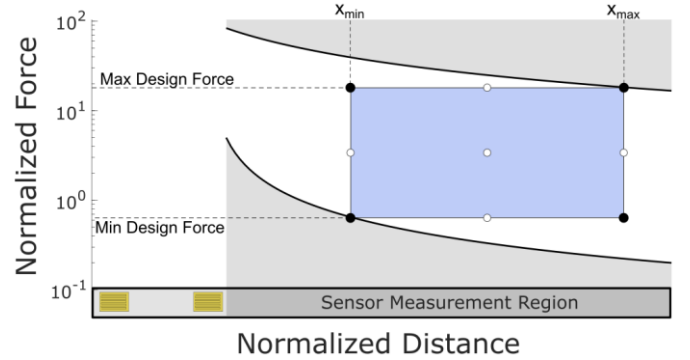


Fig. 8. Calibration points and measurement region (blue rectangle) superimposed on the full measurement region and moment isolines of Fig. 7. The recommended minimum calibration points are shown as black dots with additional calibration points shown as white dots.

B. Gauge and Measurement Region Positioning

The sensor's cross-section can be sized to accommodate various loads. This effectively shifts both isolines in Fig. 7 in the vertical direction. The relative load range (i.e., the ratio of F_{max} to F_{min}), depends primarily upon the length of the measurement region, the position of the measurement region relative to the strain gauges, and the spacing between strain gauges. In this section we will derive the relationship between these quantities.

The typical ratio of maximum allowable to minimum detectable strains for commercial load cells is approximately 100. Let this ratio be referred to as R_ϵ . Because strain is proportional to moment, the ratio M_{max}/M_{min} is also equal to R_ϵ . Using (18) and (20), we can find the ratio of the maximum measurable force to the minimum detectable force in terms of R_ϵ as follows:

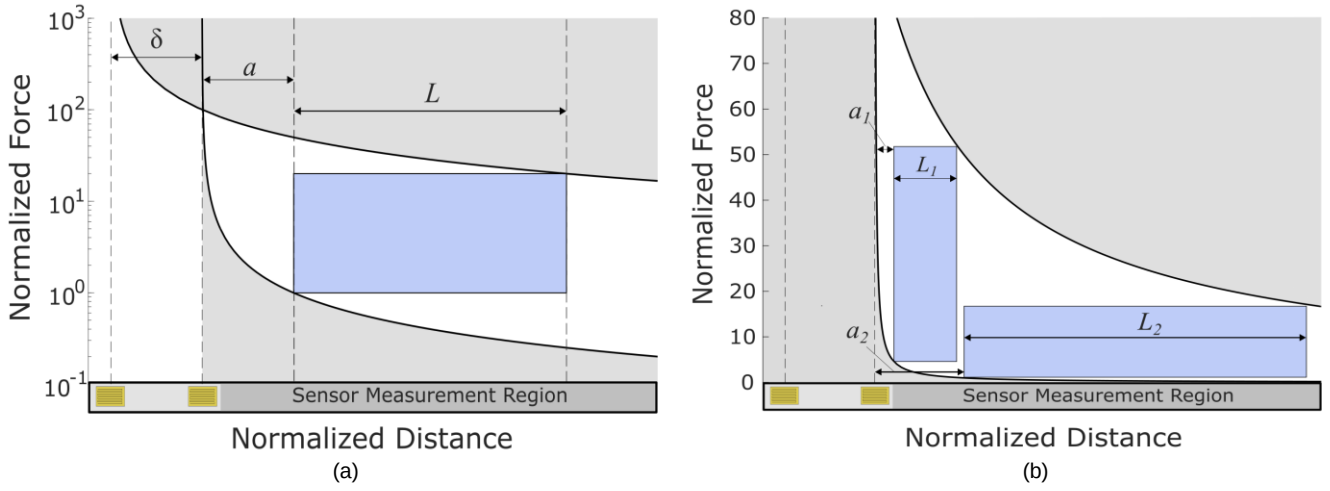


Fig. 9. Diagrams showing the variables δ , a and L from (21). (a) Variables defined. (b) A linearly scaled diagram with two sensors having different designs.

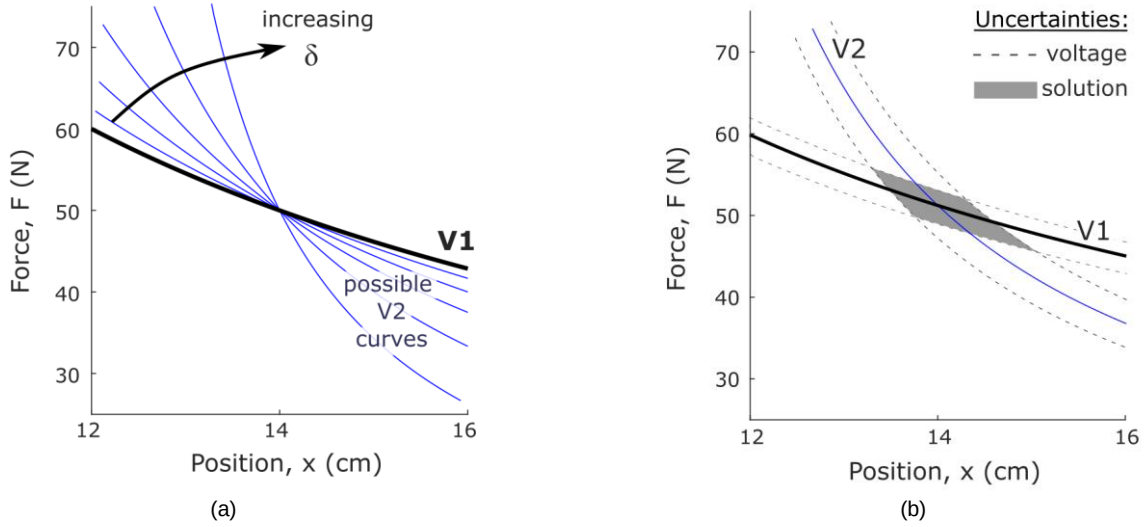


Fig. 10. (a) The intersection angle between voltage curves increases with the spacing between strain gauges. (b) The solution uncertainty region is the overlap between voltage uncertainty curves (exaggerated for illustrative purposes).

$$R_F = \frac{F_{max}}{F_{min}} = \frac{M_{max}}{M_{min}} \frac{x_{min} - d_2}{x_{max} - d_1} = R_\epsilon \frac{a}{a + L + \delta} \quad (21)$$

Here δ is the spacing between strain gauges, a is the spacing between the second strain gauge and the beginning of the measurement region, and L is the length of the measurement region.

Inspecting (21) reveals the following insights. First, the variable d_1 has no effect on the ratio R_F . However, increasing d_1 will increase the total deflection of the beam. Thus, to minimize beam deflection, the first strain gauge placement d_1 should be minimized.

Second, the distances a , L , and δ can all be used to affect the force range ratio. The force ratio is most effectively increased by decreasing L or δ . As shown in Fig. 9 (b), a measurement region with small a and L will have a high range of forces but a small range of positions. In contrast, a design with a larger a and L will have a broad positional measurement region but

lower range of measurable forces. In practice, values for δ , a , and L should be chosen to achieve the desired measurement ranges.

C. Voltage Disparity

The force-position sensor operates on the principle of measuring different voltages from each strain gauge, and the voltages measured are proportional to the moments each gauge experiences. Thus, the accuracy of the sensor's predictions is contingent on the disparity between the two strain gauges, which is proportional to δ , the distance between the strain gauges.

This can be visualized by examining isolines for each strain gauge on a position-force plane. For each gauge, a region of uncertainty on either side of the isoline indicates electronic noise. If the two strain gauges were placed at the same x -location, they would share identical voltage curves, making it impossible to solve for x and F . Fig. 10 (a) illustrates the effect of strain gauge spacing. As the distance δ between strain gauges increases, the two voltage curves become more distinct and the

angle of intersection increases. Furthermore, note that the voltage curves in Fig. (10 (a)) are voltage isolines. As the lines become more distinct from each other, the difference between voltages produced by the two sensors increases.

Fig. 10 (b) illustrates the relationship between voltage uncertainty, solution uncertainty, and angle of intersection. The solution uncertainty region is inversely related to the angle of intersection between the two voltage isolines. When the angle of intersection is small, the uncertainty region will be large and vice versa. In contrast, a larger distance between strain gauges will result in a lower solution uncertainty region.

In most cases, the constants k_1 and k_2 are nearly identical and the offset constants c_1 and c_2 are close to zero. If we neglect the variation in these constants caused by bridge balancing effects, we gain insights into the relationship between solution uncertainty and sensor design parameters. Neglecting these minor effects, we assume $k_1 = k_2 = k$, and $c_1 = c_2 = 0$. The difference in voltages detected by the two strain gauges is then found to depend upon k , F , and δ :

$$V_1 - V_2 = kF(d_2 - d_1) = kF\delta \quad (22)$$

Substituting the terms that constitute k from (7) and collecting terms allows us to assess the sources of uncertainty by conceptual groupings: (23)

$$V_1 - V_2 = \frac{1}{4}(V_s g)(Gn\delta) \frac{1}{ES} \quad (23)$$

(23) shows that the voltage disparity is influenced by electronics design factors, strain gauge configuration, beam geometry, and the beam's material properties. The electronic factors include the excitation voltage V_s , and the signal amplification, g . The strain gauge factors include gauge factor (G), the number of strain gauges used in each Wheatstone bridge (n), and the spacing between strain gauges (δ). The beam geometry affects the voltage disparity inversely through $S = y/I$, the beam's section modulus. Finally, the elastic modulus of the beam (E) also has an inverse effect on voltage disparity.

D. Model Limitations

As with any sensor, this force-position sensor has certain limitations. First, the sensor is designed to measure a force acting at a single point. A force applied over a wide region, or the application of multiple forces, will lead to erroneous or misleading data. Second, the force-position sensor is only capable of measuring the component of a force that acts perpendicular to the cantilever beam. If a temperature-compensating half or full bridge is used, the sensor will be insensitive to axial loads. However, if a quarter bridge is used, the axial component of applied forces will affect the measured results.

The sensor assumes linearity for both the bending strain experienced by the strain gauges in (1) and the voltage response of the strain gauges in (4). The nonlinearity in the voltage response of a Wheatstone bridge is negligible when temperature effects are compensated for, as in a half or full bridge. For strain to remain linear requires that the cantilever deformation remain

small, usually less than 10%, which must be taken into consideration when designing the sensor. This may, however, be a forgiving constraint - our testing on Prototype A, which used a thin, wide beam, resulted in a max tip deflection of 26.5% of the beam length, yet the calibrated device still yielded remarkably linear measurements. Minimizing deflection while maximizing strain can be achieved through more sophisticated cantilever designs, such as adding holes as stress concentrators, as in Prototype B.

E. Applications

This force-distance sensor could be useful in industrial automation settings where the weight and position of an object might be important information for a process. This sensor may be particularly beneficial in situations where the position of objects is difficult to control precisely (for example, in agricultural processing and agricultural robotics). Another potential application is in robotic gripping. This sensor can provide continuous feedback on both the gripping force and the position on the gripper at which an object is being gripped. Both factors are important for effective robotic gripping [26].

V. CONCLUSION

Measurement of force and position are common problems in engineering, but there are few if any sensor solutions that can provide continuous measurements of both quantities simultaneously. We have developed, modeled, analyzed, and validated a novel sensor that is capable of simultaneous measurement of both force and position. The force-position sensor consists of at least two strain gauges mounted at two positions along a cantilever beam. Three separate prototypes were constructed, calibrated, and validated. Across all three prototypes, the device proves to be highly linear, with a predictive R^2 value of over 0.98. The sensor can be designed for a wide variety of force and position measurement ranges. Having no moving parts, the sensor can be designed to be rugged and relatively easy to manufacture.

This force-position sensor is only an initial concept and has a great deal of room for improvement. Future work could be focused on optimizing the device for specific applications, such as robotic manipulators or industrial automation, and exploring design variations, such as the use of more than two Wheatstone bridges or different beam geometries. The sensor is currently limited to the measurement of the perpendicular component of forces, so efforts to expand the sensor concept to more general force and position distributions may also yield useful results.

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